1. Measure of Even Spacing for a Numerical Sequence

Mathematical Formulation

Inputs: A numerical sequence $V = (v_1, v_2, \ldots, v_n)$ of n real numbers.

Preprocessing and Special Cases:

1. If V cannot be converted to floats or contains NaN, score is undefined.
2. If $n = 0$, score is undefined.
3. If $n = 1$, score $S_V = 1.0$.

Core Calculation (for $n \geq 2$):

1. **Sorting:** $V'_{sorted} = (v'_{(1)}, \ldots, v'_{(n)})$.
2. **Difference Sequence (Δ):** $\Delta = (\delta_1, \ldots, \delta_{n-1})$, where $\delta_j = v'_{(j+1)} - v'_{(j)} \geq 0$.
3. **Statistical Measures of Δ (for $n \geq 2$):**
   - Mean of Differences (μ_Δ): $\mu_\Delta = \frac{1}{n-1} \sum_{j=1}^{n-1} \delta_j$.
   - Population Standard Deviation of Differences (σ_Δ): $\sigma_\Delta = \sqrt{\frac{1}{n-1} \sum_{j=1}^{n-1} (\delta_j - \mu_\Delta)^2}$. (If $n = 2$, $\sigma_\Delta = 0$).
   - Maximum Difference (δ_{max}): $\delta_{max} = \max(\delta_1, \ldots, \delta_{n-1})$.
4. **Score Components (for $n \geq 2$):**
   - Standard Deviation Score (S_σ): $S_\sigma = \frac{1}{1+\sigma_\Delta}$. (Undefined if σ_Δ is NaN/Inf).
   - Mean Spacing Score (S_μ): $S_\mu = \begin{cases} \frac{\mu_\Delta}{\delta_{max}} & \text{if } \delta_{max} \neq 0 \\ 0 & \text{if } \delta_{max} = 0 \end{cases}$.
5. **Final Vector Evenness Score (S_V for $n \geq 2$):** $S_V = S_\sigma \times S_\mu$.

2. Comparative Spread Objective for Labeled Feature Sets

Mathematical Formulation

Inputs:

- $X = (x_1, \ldots, x_M)$: Primary numerical sequence.

- $W = (w_1, \ldots, w_M)$: Secondary numerical sequence.
- $L = (l_1, \ldots, l_M)$: Binary class labels $l_i \in \{0, 1\}$.

Sub-sequence Extraction:

- $V_{X_0} = \{x_i \mid l_i = 0\}$.
- $V_{W_1} = \{w_i \mid l_i = 1\}$.

Spread Calculations:

- $S_{X_0} = \text{CalculateVectorEvennessScore}(V_{X_0})$
- $S_{W_1} = \text{CalculateVectorEvennessScore}(V_{W_1})$.

Objective Function Value (J_O): If S_{X_0} or S_{W_1} is undefined, J_O is undefined. Otherwise:

$$J_O = \frac{S_{X_0} + S_{W_1}}{2} - |S_{X_0} - S_{W_1}|$$